FairRecruit: Machine Learning Algorithm Performance Report

Empirical Benchmark Evaluation, Dimensionality Reduction Trade-offs, Influencing Factors & Decision Rationale

Project: Data Mining Project 5 • **Evaluation Split:** Stratified Validation Partition (3,832 records) • **Training Split:** 11,494 records (5-Fold Stratified CV) • **Baseline Majority:** 75.05%

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1. Executive Summary & Evaluation Framework

In algorithmic hiring and applicant shortlisting, selecting a classification algorithm requires balancing **raw predictive performance** (ranking candidates with genuine qualification/interest) against **interpretability, algorithmic fairness, and legal defensibility**. To rigorously compare model families, FairRecruit implements a zero-data-leakage testing harness: imputation, standard scaling, and one-hot encoding are fit *exclusively* on training records, and every cross-validation fold refits its feature pipelines independently.

Methodological Rigor & Data Leakage Elimination

A common flaw in academic ML submissions is calculating scalers and imputers across the entire dataset prior to splitting. In this audit, all 5 model families were evaluated on the **20% validation partition (3,832 candidates)** with identical feature spaces (10 raw predictors expanded to 45 one-hot encoded dimensions). Cross-validation scores reflect 5-fold stratified cross-validation on the 60% training partition (11,494 rows), where each fold executes its own independent pipeline fit.

2. Empirical Benchmark Evaluation Across 5 Model Families

All five algorithms were evaluated on the held-out validation cohort under fixed 0.50 probability thresholding. The majority-class zero-rule baseline is **75.05%** (predicting no candidate qualifies for shortlisting).

Model Family	Category	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC (AP)	5-Fold CV Spread
Decision Tree	Tree (Interpretable)	**77.53%**	54.90%	55.65%	**55.27%**	**0.7760**	0.4889	79.43% +/- 1.41%
Random Forest	Ensemble Bagging	76.57%	56.36%	26.88%	36.40%	0.7795	0.5184	77.69% +/- 0.44%
Logistic Regression	Linear (L2)	76.15%	54.38%	27.30%	36.35%	0.7635	0.4895	77.49% +/- 0.48%
K-Nearest Neighbors	Instance-Based (k=15)	75.00%	49.87%	41.11%	45.07%	0.7206	0.4370	76.18% +/- 0.35%
Gaussian Naive Bayes	Probabilistic	75.05%	N/A*	0.00%	0.00%	0.6280	0.3597	75.07% +/- 0.02%
Majority Baseline	Zero-Rule Threshold	75.05%	N/A	0.00%	0.00%	0.5000	0.2495	75.05% +/- 0.00%

* Gaussian Naive Bayes predicted 0 positive candidates at the 0.50 threshold (TP=0, FP=0), making precision mathematically undefined. The 75.05% accuracy is completely trivial (identical to predicting all 0s).

3. Confusion Matrix Breakdown & Prediction Behavior

Examining raw classification outcomes on the 3,832 validation applicants (956 actual positives, 2,876 actual negatives) reveals stark differences in how each algorithm approaches shortlisting:

Model	True Neg (TN)	False Pos (FP)	False Neg (FN)	True Pos (TP)	Balanced Acc	Clinical Behavior Summary
Decision Tree	2,439	437	424	532	**70.23%**	**Balanced shortlister:** Recovers 55.7% of positives with controlled FP rate (15.2%).
Random Forest	2,678	198	699	257	**59.98%**	**High specificity:** Conservative shortlister; low FP rate but misses 73.1% of positives.

Logistic Regression	2,657	219	695	261	**59.84%**	**Linear threshold:** Similar to RF, misses 72.7% of positive cohort at default 0.50.
K-Nearest Neighbors	2,481	395	563	393	**63.69%**	**Moderate recovery:** Recovers 41.1% of positives; suffers from distance degradation.
Gaussian Naive Bayes	2,876	0	956	0	**50.00%**	**Complete collapse:** Predicts 0 candidates; identical to trivial majority baseline.

4. Feature Selection vs. PCA Dimensionality Reduction

To evaluate whether dimensionality reduction improves generalization or aids algorithmic transparency, we benchmarked the Decision Tree architecture across three feature representations on the validation split:

Representation Technique	Input Dim	Accuracy	Precision	Recall	ROC-AUC	Practical Impact on Hiring Audits
All Raw Predictors	45	77.53%	54.90%	55.65%	0.7760	Preserves full feature fidelity; complete explainability down to source columns.
Importance-Based Selection	10	77.43%	54.72%	55.23%	0.7714	Preserves 99.8% of performance using 10 key features; retains input names.
PCA Orthogonal Components	10	75.37%	51.09%	29.39%	0.7043	Severe degradation: Recall drops to 29.4%; destroys audit explainability.

Critical Data Mining Finding: Why PCA Fails in HR Categorical Pipelines

Why PCA harms performance: The 10 PCA components account for only 43.25% of total variance. PCA relies on orthogonal linear combinations of continuous variables. Because the HR feature matrix consists largely of sparse one-hot dummy variables (company type, education level, major), projecting discrete 0/1 indicators into continuous eigenvectors creates noisy coordinates that destroy the sharp, axis-aligned decision boundaries required by decision trees.
Audit implication: A candidate cannot be given an explanation like 'You were rejected because Component 3 was -0.42'. PCA obliterates legal transparency.

5. Deep Dive: Key Factors Influencing Model Performance

Why do these algorithms perform so differently on the exact same dataset? Five structural factors govern their behavior:

Factor 1: Severe Class Imbalance (75% Class 0 vs. 25% Class 1)

In HR candidate screening, highly qualified candidates shortlisted for interviews represent a selective minority (24.95%). This imbalance creates a dangerous *majority baseline trap*: a trivial model predicting 'No' for every candidate achieves 75.05% accuracy. Linear models (Logistic Regression) and Naive Bayes struggle with this imbalance under a default 0.50 threshold, predicting overly conservative probabilities. Decision trees handle this significantly better by isolating pure leaf nodes based on key continuous splits.

Factor 2: Non-Linear Step Functions vs. Linear Hyperplanes

The single most dominant predictor in the dataset is city_development_index. Candidates residing in cities with an index ≤ 0.62 have a significantly higher rate of meeting qualification criteria (>55%), while candidates in high-index cities (≥ 0.85) qualify at lower rates (<15%). This is an abrupt non-linear step function. Tree-based models (Decision Tree, Random Forest) model this with a single orthogonal split. Logistic Regression must fit a smooth sigmoid curve across the entire feature space, blunting the threshold effect and dampening sensitivity.

Factor 3: High-Dimensional Categorical Sparsity (45 Encoded Features)

Expanding 7 categorical variables into one-hot dummy columns creates a 45-dimensional feature space. This triggers the **Curse of Dimensionality** for distance-based models like K-Nearest Neighbors (KNN). In 45 dimensions, Euclidean distances between points concentrate (all candidates appear equidistant), so KNN's neighborhood voting is overwhelmed by the 75% majority class, degrading precision to 49.87% and recall to 41.11%.

Factor 4: Violation of Conditional Independence (Why Naive Bayes Collapses)

Gaussian Naive Bayes assumes all features are conditionally independent given the class label. In recruitment data, features are heavily correlated (e.g., education_level is correlated with enrolled_university, and experience correlates with company_size and city_development_index). Multiplying redundant probabilities double-counts evidence, pushing posterior probabilities toward extreme 0 or 1 boundaries, causing Naive Bayes to predict 0 positive cases at the 0.50 cutoff.

Factor 5: Feature Multicollinearity and Proxies

Even when sensitive attributes like Gender, Age, and Institution are excluded from predictors, proxy correlations exist (e.g., city_development_index has an NMI of ~0.30 with certain demographic distributions). Ensemble models like Random Forest distribute split criteria across proxy features, making bias auditing harder. A single Decision Tree makes proxy usage completely transparent by exposing the exact tree nodes.

6. Algorithmic Reasoning: Which Performs Better and Why

A superficial evaluation might look solely at Random Forest's slightly higher ROC-AUC (0.7795 vs 0.7760) and conclude it is superior. However, rigorous data science engineering requires evaluating models against **both performance and operational requirements**.

The Comparative Ranking Summary:

- **Rank 1 (Overall Operational Winner): Decision Tree.** Achieves the highest validation accuracy (77.53%), highest F1-score (55.27%), highest balanced accuracy (70.23%), and strong ROC-AUC (0.7760) while providing **100% faithful explainability**.
- **Rank 2 (Best Ranking Discrimination): Random Forest.** Wins on ROC-AUC (0.7795) and Average Precision (0.5184), but suffers from low recall at default thresholds (26.88%) and acts as an uninterpretable black box.
- **Rank 3 (Linear Baseline): Logistic Regression.** Competitive ROC-AUC (0.7635) and well-calibrated probabilities, but misses non-linear step thresholds.
- **Rank 4: K-Nearest Neighbors.** Suffers from high-dimensional distance concentration; high inference latency.
- **Rank 5 (Unusable Failure Mode): Gaussian Naive Bayes.** Complete collapse on imbalanced, correlated tabular data; identical to majority rule.

Why Decision Tree is the Primary Operational Benchmark in FairRecruit

1. Regulatory Compliance (EU AI Act & EEOC Guidelines): Automated employment decision tools (AEDTs) require defensible, candidate-specific explanations. A candidate shortlisted or rejected by a Decision Tree can be provided with their exact decision path (e.g., *'Applicant 11600: city_development_index ≤ 0.62 AND experience > 4.0'*).
2. Superior F1-Score & Balanced Recall: At the operational 0.50 threshold, Random Forest and Logistic Regression miss over 73% of qualified candidates (recall < 27.3%), whereas Decision Tree recovers 55.65% of positives with an F1 of 55.27%.
3. Controlled Feature Ablation: Testing bias interventions (dropping city_development_index or proxies) is deterministic and directly auditable in a single tree, whereas ensemble feature interactions obscure the source of disparity reduction.

7. Master Viva Examination Defense Questions & Script

Q1: 'Random Forest has a higher ROC-AUC (0.7795 vs 0.7760). Why did you choose Decision Tree as your primary classifier?'

Examiner Defense: 'While Random Forest edges out Decision Tree by 0.0035 in ROC-AUC, in an automated recruitment system, ROC-AUC measures only ranking discrimination across all theoretical thresholds. In practice, hiring decisions operate at concrete thresholds. At the standard 0.50 threshold, Random Forest's recall is only 26.88% (F1: 36.40%), missing nearly three-quarters of positive applicants, whereas Decision Tree achieves 55.65% recall and 55.27% F1. Most crucially, hiring algorithms are classified as high-risk under the EU AI Act. Random Forest combines 100 trees with random subspace sampling, creating a black box. Decision Tree offers 100% faithful explainability, allowing candidates and legal auditors to inspect the exact rules governing selection.'

Q2: 'Why did Gaussian Naive Bayes achieve 75.05% accuracy with 0% precision and recall?'

Examiner Defense: '75.05% is the majority-class baseline of our dataset. In our training split, 75% of candidates are labeled 0. Gaussian Naive Bayes assumes all features are conditionally independent and normally distributed. Because our feature matrix contains 45 correlated one-hot dummy variables, the conditional independence assumption is catastrophically violated. Redundant probability multiplication forced all posterior positive probabilities below 0.50, meaning Naive Bayes predicted 0 for every single applicant. It appears accurate only because the negative class dominates the dataset.'

Q3: 'Why did PCA dimensionality reduction decrease accuracy from 77.53% to 75.37% and recall to 29.39%?'

Examiner Defense: 'PCA is a linear orthogonal transformation designed for continuous, Gaussian-distributed data. Our data consists of 10 continuous and 35 sparse categorical indicators. 10 PCA components captured only 43.25% of total variance. Projecting discrete 0/1 indicator categories into continuous eigenvectors introduced high noise and blurred the sharp axis-aligned decision boundaries that decision trees excel at finding. Furthermore, PCA destroys audit provenance, making legal compliance impossible.'